

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Summative Assessment 2

Summative Assessment (SA-2) - Answer Key

Class: Class 9

Assessment Type: Summative Assessment (SA-2)

Total Marks: 40

Duration: 2 hours

Topics Covered: Algorithms, Cyber Security, Emerging Technologies, Computational Thinking

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Part A: MCQ (10 x 1 = 10 marks)

1. **Answer:** c) $O(n^2)$
 - **Explanation:** Bubble sort compares adjacent elements and swaps them, resulting in $O(n^2)$ time complexity in the worst case.
2. **Answer:** c) Ambiguity
 - **Explanation:** A good algorithm must be unambiguous. Ambiguity violates the definiteness characteristic of algorithms.
3. **Answer:** b) A social engineering attack to steal credentials
 - **Explanation:** Phishing uses deceptive emails/websites to trick users into revealing passwords, credit card details, or other sensitive information.
4. **Answer:** c) Blockchain
 - **Explanation:** Blockchain is the underlying technology of Bitcoin. It provides a decentralised, immutable ledger for transactions.
5. **Answer:** b) To filter network traffic and block threats
 - **Explanation:** A firewall monitors incoming and outgoing network traffic and blocks malicious traffic based on security rules.
6. **Answer:** c) Parallelogram
 - **Explanation:** In flowcharts, a parallelogram represents input/output operations. Rectangle = process, Diamond = decision, Oval = start/end.
7. **Answer:** b) An email asking for password pretending to be from a bank
 - **Explanation:** This is a classic example of phishing - impersonating a trusted entity to steal credentials.
8. **Answer:** b) A subset of AI that enables systems to learn from data
 - **Explanation:** Machine Learning allows computers to learn patterns from data and make decisions without being explicitly programmed.
9. **Answer:** b) Symmetric encryption
 - **Explanation:** Symmetric encryption uses the same key for both encryption and decryption (e.g., AES). Asymmetric uses different keys (public/private).
10. **Answer:** b) Internet of Things
 - **Explanation:** IoT refers to the network of physical devices embedded with sensors, software, and connectivity to exchange data.

Part B: Short Answer (8 x 2 = 16 marks)

1. Write an algorithm to find the second smallest element in an unsorted list.
 - **Answer:**

ALGORITHM SecondSmallest(list)
INPUT: An unsorted list of n elements
OUTPUT: The second smallest element

```
1. IF n < 2 THEN RETURN "List too small"
2. SET smallest = list[0]
3. SET second = infinity
4. FOR i = 1 TO n-1 DO
    a. IF list[i] < smallest THEN
        second = smallest
        smallest = list[i]
    b. ELSE IF list[i] < second AND list[i] != smallest THEN
        second = list[i]
5. RETURN second
```

- Time complexity: $O(n)$ - single pass through the list.
- **Marking:** 1 mark for algorithm, 1 mark for time complexity analysis.

2. **Differentiate between encryption and decryption.**

- **Answer:**
 - Encryption: Converting plaintext into ciphertext using an algorithm and key. Example: Encrypting a message before sending via email.
 - Decryption: Converting ciphertext back into plaintext using the appropriate key. Example: Receiving and reading the encrypted email.
- **Marking:** 1 mark for each definition with example.

3. **Explain three types of malware.**

- **Answer:**
 - Virus: Attaches to legitimate files, spreads when executed. Prevention: Use antivirus software.
 - Worm: Self-replicating, spreads across networks without user interaction. Prevention: Keep software updated, use firewalls.
 - Ransomware: Encrypts files and demands payment for decryption. Prevention: Regular backups, avoid suspicious links.
- **Marking:** 1 mark for any two types with prevention, 1 mark for the third.

4. **What is Big Data? Explain the 5 V's.**

- **Answer:** Big Data refers to extremely large datasets that cannot be processed by traditional tools.
 - Volume: Massive amount of data generated.
 - Velocity: Speed at which data is generated and processed.
 - Variety: Different formats (structured, unstructured, semi-structured).
 - Veracity: Accuracy and trustworthiness of data.
 - Value: Useful insights extracted from data.
- **Marking:** 1 mark for definition, 1 mark for any four V's correctly explained.

5. **Write a Python program to implement selection sort.**

```
• Answer:

def selection_sort(arr):
    n = len(arr)
    for i in range(n):
        min_idx = i
        for j in range(i+1, n):
            if arr[j] < arr[min_idx]:
                min_idx = j
```

```

        arr[i], arr[min_idx] = arr[min_idx], arr[i]
    return arr

```

```

data = [64, 25, 12, 22, 11]
print("Original:", data)
print("Sorted:", selection_sort(data))
# Trace: [64,25,12,22,11] -> [11,25,12,22,64] -> [11,12,25,22,64] ->
[11,12,22,25,64] -> [11,12,22,25,64]

```

- **Marking:** 1 mark for correct algorithm, 1 mark for trace on given input.

6. Explain two-factor authentication.

- **Answer:** Two-factor authentication (2FA) requires two different forms of verification: something you know (password) and something you have (phone/OTP) or something you are (biometric). It is more secure because even if the password is compromised, the attacker still needs the second factor.
- **Marking:** 1 mark for explanation of 2FA, 1 mark for why it's more secure.

7. What is cloud computing? Differentiate public, private, and hybrid cloud.

- **Answer:** Cloud computing delivers computing services (servers, storage, databases) over the internet.
 - Public Cloud: Owned by third-party providers (e.g., AWS, Google Cloud). Shared infrastructure.
 - Private Cloud: Dedicated to a single organisation. More control and security.
 - Hybrid Cloud: Combines public and private clouds. Allows data and applications to move between them.
- **Marking:** 1 mark for cloud computing definition, 1 mark for differentiation with examples.

8. Write pseudocode for binary search and trace it.

- **Answer:**

```

ALGORITHM BinarySearch(arr, target)
INPUT: Sorted array arr, target value
OUTPUT: Index of target or -1

```

1. SET low = 0, high = length(arr) - 1
 2. WHILE low <= high DO
 - a. SET mid = (low + high) / 2
 - b. IF arr[mid] == target THEN RETURN mid
 - c. ELSE IF arr[mid] < target THEN low = mid + 1
 - d. ELSE high = mid - 1
 3. RETURN -1
- Trace for target 23: low=0,high=9,mid=4(16<23) -> low=5,high=9,mid=7(56>23) -> low=5,high=6,mid=5(23==23) -> Found at index 5.
 - **Marking:** 1 mark for pseudocode, 1 mark for correct trace.

Part C: Long Answer (4 x 3 = 12 marks)

1. Write a Merge Sort program and trace it.

- **Answer:** Key components:
 - Divide array into halves recursively until single elements
 - Merge sorted halves in correct order
 - Trace: [38,27,43,3,9,82,10] -> divide -> merge steps
 - Time complexity: $O(n \log n)$ for best, average, and worst cases

- **Marking:** 1 mark for algorithm, 1 mark for trace, 1 mark for time complexity analysis.
2. **Cyber Safety Policy for a School.**
- **Answer:** Must include:
 - Five threats: Phishing, Malware, Cyberbullying, Identity theft, Password attacks
 - Prevention for each: Education, software updates, reporting mechanisms, strong passwords, 2FA
 - Incident response: Identify, Contain, Eradicate, Recover, Lessons Learned
 - Monitoring: Content filters, activity logs, awareness programs
 - **Marking:** 1 mark for threats and prevention, 1 mark for incident response plan, 1 mark for monitoring approach.
3. **Explain AI, IoT, and Blockchain with examples.**
- **Answer:**
 - AI: Systems that mimic human intelligence. Applications: Smart tutors, automated grading. Ethical concerns: Bias, privacy, job displacement.
 - IoT: Network of connected physical devices. Components: Sensors, actuators, gateway, cloud. Smart home example: Smart thermostat adjusting temperature automatically.
 - Blockchain: Distributed ledger with blocks linked by cryptography. Verification: Consensus mechanisms (proof of work/stake). Beyond crypto: Supply chain tracking, digital certificates.
 - **Marking:** 1 mark for each technology (definition, example, and one additional point).
4. **Design a Sorting Algorithm Visualiser project.**
- **Answer:** Key components:
 - Problem statement: Visualise sorting algorithms step by step for educational purposes
 - Data structure: Array stored as list, steps stored as list of states
 - Algorithms: Bubble sort (compare adjacent), Insertion sort (shift elements), Selection sort (find minimum)
 - Web interface: JavaScript animations using `setTimeout()`, colour coding for comparisons/swaps
 - **Marking:** 1 mark for problem statement and data design, 1 mark for algorithm descriptions, 1 mark for web visualisation approach.

Part D: Application Based (2 x 1 = 2 marks)

1. **Social Media Data Protection.**
- **Answer:**
 - Encryption: End-to-end encryption for messages (e.g., Signal protocol)
 - Firewalls/IDS: Monitor server traffic, detect DDoS attacks
 - Password policies: Minimum length, complexity, 2FA enforcement
 - Regulations: GDPR, IT Act compliance, data minimisation principles
 - **Marking:** 1 mark for technical measures, 1 mark for policies and regulations.
2. **Healthcare Digitisation.**
- **Answer:**
 - Cloud: HIPAA-compliant storage, scalable for medical records
 - IoT: Wearable devices monitoring heart rate, sending data to doctors
 - Cyber security: Encryption at rest and in transit, access controls, audit logs
 - Ethical considerations: Patient consent, data ownership, right to be forgotten
 - **Marking:** 1 mark for cloud and IoT applications, 1 mark for security and ethics.